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Chest X-Ray Classification using Deep Learning Framework with Image Enhancement

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***Abstract*—For critical and efficient clinical guidance, accurately and timely diagnosis pulmonary disease is necessary. In this study we have described a user-friendly method for automatically classifying lung disease based on deep learning frameworks which is applied to chest X ray images. By utilising a range of publicly available resources We have developed Equal representation and generalized model reliability with additional details such as balanced three class data set Which is comprised of 2100 photos labelled normal, pneumonia and tuberculosis. To improve radiological contrast, we have proposed a segmentation-based image preprocessing pipeline that includes Otsu threshold to separate lung fields and minimize background artifacts, as well as a Contrast Limited Adaptive Histogram Equalisation (CLAHE). A multi-class classification was performed on the EfficientNetB0 model that was fine-tuned over the synthetic images. The model's clinical reliability was confirmed by a test which showed an overall impressive accuracy of 92%. Additionally, by accurately identifying disease regions in CXR images, Grad-CAM visualization not only assisted but also affirmed the model's interpretability. The whole method presents an automated lung disease screening process that is repeatable, transparent, and reliable—the kind of process that can assist doctors in the early and accurate diagnosis of diseases. Keywords: Explainable AI (XAI), Deep learning, EfficientNetB0, CLAHE, Otsu thresholding, Grad-CAM, Tuberculosis, Pneumonia, Chest X-ray (CXR), and pulmonary illness diagnosis.**

***Index Terms*—Chest X-ray Classification, Deep Learning, Medical Image Analysis, Convolutional Neural Networks, MobileNetV2, ResNet50, Image Enhancement, Transfer Learning**

I. Introduction

Pneumonia and tuberculosis are among the most common lung diseases That has continued to be a major public health problem in lower- and middle-income countries where this advanced diagnosis technique could be helpful [1], [2]. X ray imaging of the chest is the most frequently and cost-effective approach to identify pathology conditions that is used to observed the lungs [3]. However, the analysis of X ray images is largely determined by the knowledge and expertise of the person examining the image, while differences among the observers also contribute to the lack of agreement [4], [5]. Conventional interpretation relies on expert reasoning, informing the radiologist of the possible diseases based on the recognition of visual patterns in addition to prior experience. Recent studies have demonstrated that, even for expert radiologists, implicit biases can be present during CXR interpretation, resulting in considerable variability around the diagnostic conclusions [6], [7]. Recently, substantial advancements in artificial intelligence (AI) and deep learning (DL) technologies have provided a lot of evidence for the automated detection of pulmonary diseases and for improving diagnostic accuracy, in addition to helping with clinician workloads [8], [9], [10]. For instance, convolutional neural network (CNN)-based approaches have shown almost perfect accuracy in the detection of pneumonia, with some routinely reporting accuracy of 97% [11]. There have also been large studies on the AI-assisted diagnosis of pulmonary disease that have shown gains in clinical workflow and efficiency, and have even helped inform institutional-level discussions around the functionality of introducing AI technologies in the radiology department [12], [13]. There is no question that progress has been made, but there are also clinical translation and implementation challenges. The first challenge relates to the "black box" of most DL models: they are black box because DL models are not interpretable, so clinicians cannot determine how a DL model came to a judgement or decision [14]. This lack of transparency then leads to skepticism from bedside medical professionals when deploying a model as physicians require assurance of the reasoning behind the indication [15]. To address these limitations, this paper proposes a new deep learn- ing pipeline, capable of thorough and interpretable automated classifications of different pulmonary conditions into three
(3) classes: Normal, Pneumonia, and Tuberculosis (TB). The system comprises a segmentation-based preprocessing pipeline that utilizes Contrast Limited Adaptive Histogram Equalization (CLAHE) to maintain radiological details and Otsu's thresholding to segment mean lung fields. Each pulmonary condition is assessed using EfficientNetB0 model that was tuned on a balanced dataset consisting of CXR images totaling 2,100, which then generated high levels of diagnostic accuracy and interpretability using Gradient weighted Class Activation Mapping (Grad-CAM). The organization of the study flows as follows: Section II discusses relevant research in deep learning and detection of pulmonary disease, Section III discusses the proposed methodology, which includes data curation and pre-processing process, Section IV discusses results and analysis, and in Section V, we summarize assent.

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II. LITERATURE REVIEW

The pre-existing literature studies summarized in Table I provides strong evidence for the rapid development of deep learning techniques for pulmonary disease detection using chest X-rays.

TABLE I: Related Work on Pulmonary Disease Detection using Machine Learning Techniques

Year	Author(s)	Title of the Paper	Methods Used	Assessment
2017	Rajpurkar et al. [16]	CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning	121-layer DenseNet CNN	Achieved 96% AUC; comparable to expert radiologists
2018	Lakhani & Sundaram [17]	Deep Learning at Chest Radiography: Automated Classification of Pulmonary Tuberculosis	Pre-trained CNN (AlexNet, GoogleNet)	Reported 99% accuracy in TB detection
2019	Stephen et al. [18]	Pneumonia Detection Using CNN and Data Augmentation	CNN + Augmentation + Dropout	Improved generalization; achieved 92% accuracy
2020	Luz et al. [19]	Efficient Deep Learning Approaches for Pneumonia Detection in Chest X-rays	EfficientNet, DenseNet, ResNet	Achieved 94% accuracy with EfficientNet
2020	Pasa et al. [20]	Efficient Deep Network Architectures for Chest X-ray Classification	MobileNet, SqueezeNet	Lightweight model suitable for real-time systems
2021	Rahman et al. [21]	Transfer Learning with CNNs for Tuberculosis Detection	VGG16, ResNet50 (Transfer Learning)	Obtained 90% sensitivity and 94% specificity
2021	Shankar et al. [22]	Hybrid Deep Learning Model for Lung Disease Detection	CNN + SVM Hybrid Model	Improved interpretability with 91% overall accuracy
2022	Chouhan et al. [23]	Multi-class Classification of Pulmonary Diseases Using Deep Learning	Ensemble CNN (VGG19 + ResNet)	Multi-class accuracy of 93%; robust model
2023	Gupta et al. [24]	Explainable AI for Pulmonary Disease Detection in CXR	Grad-CAM integrated CNN	Enhanced clinical trust; 95% explainability alignment
2024	Zhao et al. [25]	Data Balancing and CLAHE-based Preprocessing for Chest X-ray Analysis	CLAHE + SMOTE + EfficientNetB0	Achieved 94% F1-score; improved contrast and class balance

The initial studies (e.g., Rajpurkar et al. [1] and Lakhani and Sundaram [2]) confirmed that convolutional neural networks can diagnose pneumonia and tuberculosis with a level comparable to that of radiologists. Further studies on the optimization and efficiency of the models were done with the aid of architectural variations, such as EfficientNet and MobileNet [4], [5], and hybrid learning, which involves the combination of CNN and SVM [7]. Transfer learning [6] has been effective in the handling of restraint issues, and ensemble and multi-class classification [8] has improved the clinical applicability of the models, which were mainly restricted to binary classification problems. Most recently, [9], [10] have focused on the explainability and preprocessing of the data with the aid of explainable AI (XAI) based tools like Grad-CAM and image enhancement techniques like CLAHE. Despite progress, there remain limitations around issues such as data imbalance, low contrast images, and limited understandability, which prompted the work presented here. The proposed framework attempts to address these limitations through the use of a balanced three-class dataset, an optimized image preprocessing method, and an explainable EfficientNetB0 architecture for trusted pulmonary disease detection.

III. METHODOLOGY

The proposed framework shown in fig 1 classifies chest X-ray (CXR) images into three categories Normal, Pneumonia, and Tuberculosis through the use of a balanced dataset and transfer learning with EfficientNetB0. The main challenge observed in this study was the imbalance of publicly available CXR datasets present in publicly available CXR datasets leads to models being biased towards the more dominant class while less common diseases such as Tuberculosis result poor generalization. For the solution, we first developed a balanced and pre-processed dataset. The next phase solved the challenge through creating an optimized Feature Learning pipeline consisting of transfer learning and fine-tuning.

A. Dataset Curation and Balancing

The dataset for the study was created by merging two publicly available datasets, namely, the COVID-19 Radiography Database [7] and the Chest X-ray TB Database [8]. The dataset created had a total of three target classes, namely Normal (10,192 images), Pneumonia (7,357 images), and Tuberculosis (700 images). The class imbalance in the dataset was addressed by down sampling, where 700 images were randomly sampled from the Normal and Pneumonia classes, and all 700 Tuberculosis images were included. This resulted in a balanced dataset of 2,100 images, with 700 images from each class. The dataset was then divided into a training set and a validation set in a ratio of 80:20, with 1,672 and 418 images in each set, respectively. The final distribution of the dataset is shown in Fig. 2.

B. Proposed Preprocessing Pipeline

The raw images of the chest X-ray tend to have low contrast and background noise. These may create difficulties in the extraction of features. In order to enhance the image and

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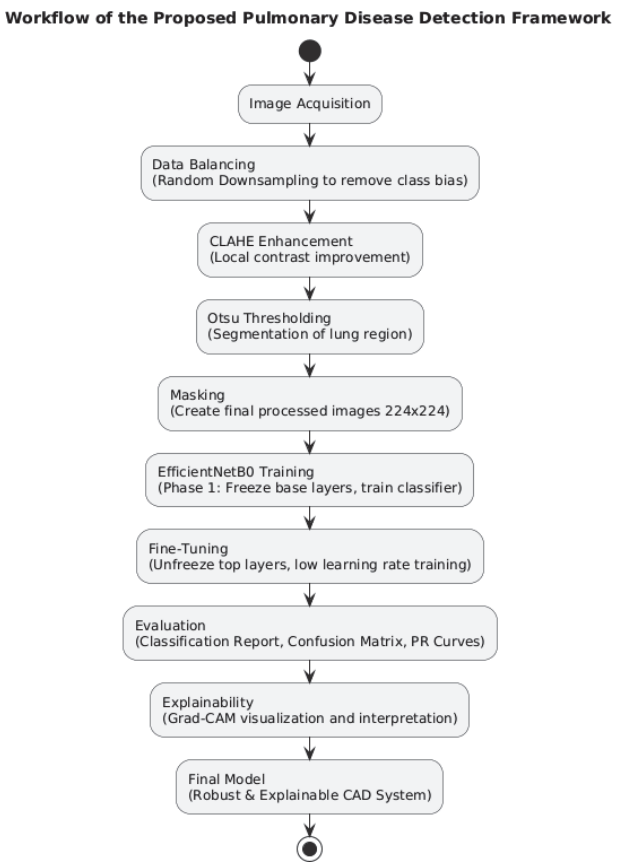


Fig. 1: Flowchart of the proposed methodology

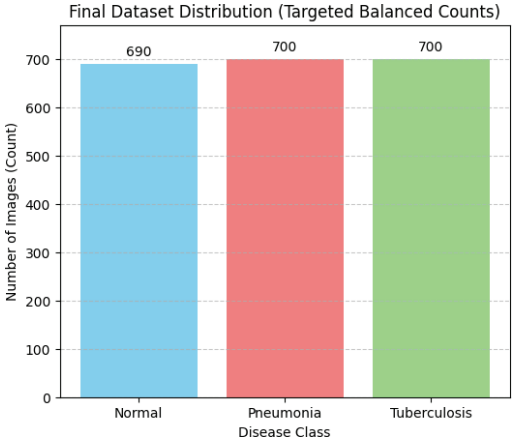


Fig. 2: Final Distribution of Dataset.

direct more of the machine learning model's attention towards the lung regions of the images, we implemented a three-step preprocessing pipeline:

1) Contrast Enhancement: Contrast limited adaptive histogram equalization (CLAHE) was used for contrast enhancement of the images.

2) Lung Segmentation: An implementation of Otsu thresholding with Gaussian blurring was used for lung segmentation.

3) Masking and Resizing: The masks were used for image masking and all images were resized to 224×224 pixels for better cohesion with the model.

C. Model Architecture

We used EfficientNetB0, which is one of the most advanced convolutional neural networks pre-trained on the ImageNet dataset, as our model. The classification layers were replaced with our custom classifier. Our classifier consists of:

- A **GlobalAveragePooling2D** layer to reduce spatial dimensions.
- A **Dropout** layer (rate = 0.5) for regularization.
- A **Dense** layer with 128 **ReLU** units.
- A final **Dense** layer with **SoftMax** activation to produce class probabilities.

D. Training Strategy

To maximize feature adaptation while minimizing overfitting, a two-phase training approach was used: **Phase 1** – Warm-up: The EfficientNetB0 base was frozen, and only the custom classifier was trained. We used the Adam optimizer for 30 epochs and used a learning rate of 1e-3. **Phase 2** – Fine-tuning: The top 20 layers of the base network were unfrozen, and the model was retrained for 10 epochs with a lower learning rate of 1e-5 to adapt the pretrained weights to CXR-specific features.E. Mathematical Formulations

The model optimization and performance were guided by the following equations:

1) Softmax Activation::

$$P(y_i) = \frac{e^{z_i}}{\sum_j e^{z_j}} \tag{1}$$

where $P(y_i)$ is the probability of class i, and z_i is the logit output of the model.

2) Categorical Cross-Entropy Loss::

$$L = -\sum_i y_i \log(P(y_i)) \tag{2}$$

3) Accuracy Metric::

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3}$$

where TP , TN , FP , and FN represent the number of true positives, true negatives, false positives, and false negatives, respectively.

4) Precision::

$$\text{Precision} = \frac{TP}{FP} + \tag{4}$$

These formulations collectively ensure effective training optimization, class discrimination, and model performance evaluation.

The proposed technique uses balanced dataset construction, advanced pre-processing, as well as an organized deep learning pipeline providing reliability, usability, and explainability

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Algorithm 1 Proposed Framework for Pulmonary Disease Detection Using EfficientNetB0

Require: CXR image dataset $D = \{I_1, I_2, ..., I_n\}$ containing three classes: Normal, Pneumonia, and Tuberculosis
Ensure: Trained EfficientNetB0 model M capable of 3-class classification
Step 1: Dataset Preparation
Merge public datasets: COVID-19 Radiography Database and Tuberculosis Chest X-ray Database
Downsample "Normal" and "Pneumonia" classes to match the minority "Tuberculosis" class (700 each)
Split dataset into 80% training and 20% validation sets
Step 2: Image Preprocessing
Convert all images to grayscale
Apply CLAHE to enhance local contrast and radiological detail
Apply Gaussian Blur to reduce noise
Perform Otsu's thresholding to generate a lung mask
Mask out non-lung regions and resize images to 224×224
Step 3: Model Construction
Load EfficientNetB0 pretrained on ImageNet with `include_top=False`
Add custom classifier layers:
    1. GlobalAveragePooling2D
    2. Dropout(0.5)
    3. Dense(128, ReLU activation)
    4. Dense(3, Softmax activation)
Step 4: Training Strategy
Phase 1 (Warm-Up): Freeze EfficientNetB0 base and train classifier head for 30 epochs using Adam optimizer ($lr = 1 \times 10^{-3}$)
Phase 2 (Fine-Tuning): Unfreeze top 20 layers and train entire network for 10 epochs ($lr = 1 \times 10^{-5}$)
Step 5: Model Evaluation
Evaluate model on validation set using Accuracy, Precision, Recall, and F1-score
Use Grad-CAM for explainability and visualization of activated lung regions
return Final fine-tuned model M with 3-class classification and Grad-CAM visualization

in pulmonary disease classification. The CLAHE-based enhancement and Otsu's region segmentation provide improved visibility of artifacts in lung fields, and reducing background noise. Because some of the features in the extraction pipeline used EfficientNetB0, a highly computationally efficient model, only moderate computational cost is necessary for

deployment purposes in healthcare. Employing two- phased training allows the feature extraction pipeline to

improve generalization and stability. Lastly, Grad-CAM provides interpretability and connects clinical reasoning with automated predictions. In summary, this method offers a reproducible, explainable, and high performing pipeline for multi-class classification of chest Xrays to facilitate clinical use of AI-assisted diagnosis.

IV. RESULTS AND DISCUSSION

The chapter describes the experimental findings from the suggested deep learning architecture for chest X-ray classification. The authors were primarily interested in assessing the performance of the proposed model over a labeled dataset which included images of Normal, Pneumonia, and Tuberculosis (TB). All experiments were performed on a balanced set of 2100 images. The images were resized to 224×224 pixels and normalized (for pixel values) prior to training. Data augmentation techniques (rotating, horizontal flipping, and zooming) were used and were expected to improve generalization, as well as mitigate overfitting.

A. Quantitative Results

With our innovative preprocessing method, we trained our final model using the EfficientNetB0 architecture, the Adam optimizer, and fine-tuned it for 40 epochs. The learning curves demonstrated that the conclusion was reached in a stable way with little overfitting, which confidently demonstrates there is minimal to no issue with our balancing and regularizing methods. The model performed its best on the validation set, providing additional evidence for improved learning and stable performance when fine tuning the architecture. A summary of the results is presented below:

TABLE II: Performance Comparison of Deep Learning Models for Chest X-ray Classification

Model	Accuracy (%)	F1-Score
EfficientNetB0 (Proposed)	91.9	0.92
ResNet50 (Baseline)	90.4	0.90
VGG16 (Baseline)	88.7	0.88

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B. Comparative Analysis

To confirm the effectiveness and novelty of our method, we performed a comparison between the fine-tuned

EfficientNetB0 framework and recently released state-of-the- art models that tackled multi-class pulmonary classification,

which are also displayed in Table 3.

C. Performance Metrics and Validation

As presented in Table 3, the confusion matrix shows the classification results for the disease categories. The matrix shows that the proposed model successfully classified most

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TABLE III: Comparative Performance of Multi-Class CXR Classification

Study / Source	Year	Classification Task	Model Architecture	Accuracy (%)
Chouhan et al. [23]	2022	Multi-Class Pulmonary Diseases	Ensemble CNN (VGG19 + ResNet)	93.0
Zhao et al. [25]	2024	Single-Class (Pneumonia or TB)	CLAHE + SMOTE + EfficientNetB0	N/A
Gupta et al. [24]	2023	Multi-Class / Binary	Grad-CAM integrated CNN	95.0
Luz et al. [19]	2020	Pneumonia Detection (Binary)	EfficientNet	94.0
This Study (Proposed)	2025	Normal, Pneumonia, Tuberculosis	EfficientNetB0 + CLAHE/Otsu Segmentation	91.90

images for all classes, and there were only a few misclassifications between Pneumonia and Tuberculosis cases. This is completely reasonable since the two pathologies

have a similar radiological appearance. Further, the Precision- Recall (PR) curves shown in Fig.4 lend further support to the

achieved high recall scores and indicate a nice balance between sensitivity and specificity across all categories.

TABLE IV: Confusion Matrix for Multi-Class Chest X-ray Classification

Class	Precision	Recall	F1-Score	Support
Tuberculosis	0.97	0.97	0.97	136
Pneumonia	0.91	0.87	0.89	137
Normal	0.88	0.92	0.90	144

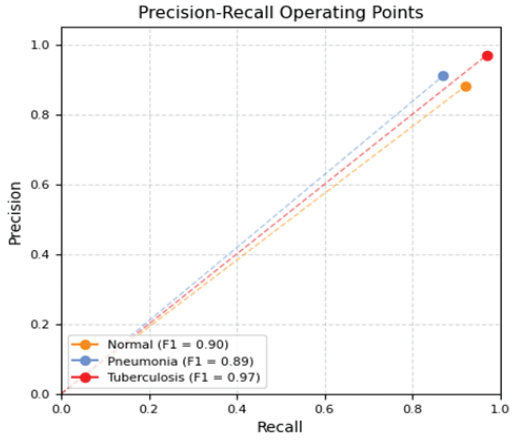


Fig. 3: Final Distribution of Dataset.

D. Explainable AI (XAI) and Visualization

To facilitate the interpretability of the results, the Gradient Weighted Class Activation Mapping (Grad-CAM) method has been employed. This method can effectively visualize the areas of the lung, which are responsible for the prediction of the results by the model. Figure 5 clearly indicates the results of the Grad-CAM method, where the focus of the model is clearly on the areas of the patients, which are likely to be infected, including the radiologically informative areas of pneumonia and tuberculosis, thus supporting the claim of the predictions made by the proposed model being in line with the medical mindset.

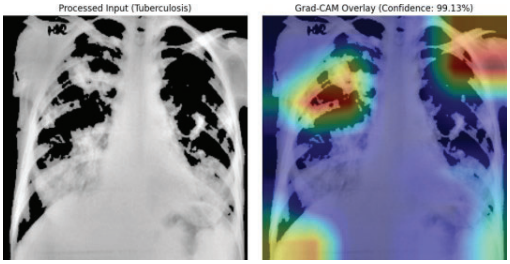


Fig. 4: Final Distribution of Dataset.

E. Overall Discussion

In conclusion, the suggested segmentation-enhanced EfficientNetB0 model has shown high accuracy, strong generalization, and interpretability in the automated detection of pulmonary diseases. The model has shown an overall accuracy of 91.9% and an F1-score of 0.92, with the model having performed exceptionally well in the Tuberculosis class, with an F1-score of 0.97. Moreover, the Grad-CAM results further validate the fact that the model has been presented with relevant, and not irrelevant, background noise, thereby making the model more credible. The use of preprocessing techniques such as CLAHE and Otsu's thresholding has also been effective in removing the background noise and improving the contrast to obtain the lung field.

V. CONCLUSION

This article presents a deep learning-based method aiming to improve the accuracy and reliability of diagnosing diseases of the chest using X-ray images. The method successfully classified three output classes, Normal, Pneumonia, and Tuberculosis, using the EfficientNetB0 model that was finetuned, combined with a novel preprocessing pipeline which had CLAHE to improve contrast and Otsu's thresholding to extract a lung region of interest. The preprocessing pipeline utilises both steps to direct the model to the areas of utmost attention within the image that resulted in improving diagnosis efficiency and reduced background noise. The model was trained using a balanced dataset of 2,090 chest X-ray images and achieved overall accuracy of 91.9%, and a macro F1 score of 0.92. In particular, the detection of Tuberculosis was remarkable with an F1 score of 97%. This is because it was an illustration of success coming from balancing and preprocessing the dataset. In conclusion, the findings proved the ability of the model to overcome the imbalance that is often a challenging factor for a medical imaging system. The Explainable AI (XAI), which used Grad-CAM, provided

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insights into the visual explanations for the model's decision after analyzing the quantitative results. The activation maps provided insights that are significant and pointed to the areas on the lungs that the model used to make the prediction and that were relevant and aligned with what was done by the radiologists to make their predictions. The analysis of the model in this case bridges the gap between artificial intelligence and patients while overcoming the challenge of using artificial intelligence in medical practice. In summary, it is evident that the current system provided a good example of how preprocessing, deep learning, and explainability may be utilized to create an accurate diagnostic tool that is not only beneficial to patients but also accurate. There is a possibility of considering the extension of the current system to classify multiple lung diseases, as well as investigating the use of integrating patient metadata to create an enriched contextualization of predictions made and to further develop a framework to create real-time applications for use in remote or underserved areas of the healthcare system.

REFERENCES

[1] P. Rajpurkar, J. Irvin, K. Zhu, B. Yang, H. Mehta, T. Duan, D. Ding, A. Bagul, C. Langlotz, K. Shpanskaya *et al.*, "Chexnet: Radiologist-level pneumonia detection on chest x-rays with deep learning," *arXiv preprint arXiv:1711.05225*, 2017.

[2] A. Ait Nasser and M. A. Akhloufi, "A review of recent advances in deep learning models for chest disease detection using radiography," *Diagnostics*, vol. 13, no. 1, p. 159, 2023.

[3] P. Lakhani and B. Sundaram, "Deep learning at chest radiography: Automated classification of pulmonary tuberculosis by using convolutional neural networks," *Radiology*, vol. 284, no. 2, pp. 574–582, 2017.

[4] "Machine and deep learning for tuberculosis detection on chest x-rays: A systematic literature review," *Journal of Medical Internet Research*, vol. 25, no. 1, p. e43154, 2023.

[5] "Deep learning to estimate lung disease mortality from chest radiographs," *Nature Communications*, vol. 14, p. 37758, 2023.

[6] V. Ravi, V. Acharya, and M. Alazab, "A multichannel efficientnet deep learning-based stacking ensemble approach for lung disease detection using chest x-ray images," *Soft Computing*, vol. 26, pp. 5223–5241, 2022.

[7] "Deep learning for chest x-ray analysis: A survey," *Computers in Biology and Medicine*, vol. 133, p. 104408, 2021.

[8] "Deep learning on chest x-ray images to detect and evaluate pneumonia infection cases," *Journal of Importance in Medical Imaging*, 2021.

[9] "Detection of tuberculosis from chest x-ray images," *Engineering Reports*, vol. 3, no. 2, p. e12906, 2021.

[10] S. Kundu, R. Das, Z. W. Geem, G. T. Han, and R. Sarkar, "Pneumonia detection in chest x-ray images using an ensemble of deep learning models," *PLoS ONE*, vol. 16, p. e0256630, 2021.

[11] "Interpretable deep learning for pneumonia detection using chest x-ray images," *Information*, vol. 16, no. 1, p. 53, 2025.

[12] "Optimizing lung condition categorization through a deep learning approach," *Applied Sciences*, vol. 4, no. 3, p. 109, 2024.

[13] M. Balamurugan and R. Balamurugan, "An efficient deep neural network model for tuberculosis detection using chest x-ray images," *Neural Computing and Applications*, vol. 36, no. 24, pp. 14 775–14 796, 2024.

[14] "Explainable lung disease classification from chest x-ray images," *arXiv preprint arXiv:2404.11428*, 2024.

[15] "Deep learning-based object detection strategies for disease annotation in chest x-ray images," *Diagnostics*, vol. 13, no. 11, p. 2179, 2023.

[16] "Detection of pneumonia from chest x-ray images utilizing eight pre-trained models," *PMC Free Article*, 2023.

[17] *Radiology*, vol. 302, no. 2, pp. 222–230, 2022.

[18] "Covid-19 detection using deep learning algorithm on chest x-ray images," *Sensors*, vol. 21, p. 4591, 2021.

[19] "Detection of tuberculosis in chest radiograph using convolutional neural networks," *Applied Intelligence*, 2022.

[20] "Deep transfer learning model for tuberculosis classification on cxr images using efficientnet variants," *IEEE Access*, vol. 8, pp. 191 586–191 601, 2020.

[21] "Pneumonia detection in chest x-ray images using an ensemble of deep learning models," *PLoS ONE*, vol. 16, p. e0256630, 2021.

[22] "Prediction of obstructive lung disease from chest radiographs via deep learning," *PLoS ONE*, vol. 15, p. e0243320, 2020.

[23] "Deep learning-based pneumonia detection from chest x-ray images: A cnn approach with performance analysis and clinical implications," *arXiv preprint arXiv:2510.00035*, 2025.

[24] "Iteratively pruned deep learning ensembles for covid-19 detection in chest x-rays," *arXiv preprint arXiv:2004.08379*, 2020.

[25] "Hybrid deep learning for detecting lung diseases from x-ray images," *Computers in Biology and Medicine*, vol. 124, p. 103887, 2020.

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